UNIT 10 · TOPIC 5

# Yield Monitors and Variable-Rate Applicators

*Syllabus topic: Yield monitors and variable rate applicators for fertilizers, seed and chemicals*

## Yield monitors

A YIELD MONITOR on a combine continuously measures:

- Grain FLOW (impact or optical/mass-flow sensor).
- Grain MOISTURE.
- Header width and ground speed.
- GPS position.

The on-board computer calculates yield per unit area at each point; over the field this produces a **YIELD MAP** — the primary record of spatial variability and the starting point of the PA cycle.

- **Essential for accuracy:** calibration against weighed loads, and correction for FLOW DELAY (time lag through the combine).
- Yield maps reveal consistently high/low zones for targeted management.

## Variable-rate applicators

Variable-rate applicators adjust the rate of seed, fertiliser or chemical across the field instead of applying a fixed rate.

| Equipment | What varies |
|---|---|
| Variable-rate seed drills / planters | Seed rate / spacing |
| Fertiliser applicators (spinner/pneumatic) | Metering rate |
| Sprayers | Rate, nozzle and section control |

How the rate is set (map-based vs sensor-based) and the control electronics: Topic 9.

## Practice MCQs

**Q41. A yield monitor on a combine measures grain flow together with:**

(a) Moisture, header width, speed and GPS position (b) Air pressure
(c) Soil pH (d) Sound level

**Q42. The primary record of within-field variability is the:**

(a) Road map (b) Yield map
(c) Contour map (d) Cadastral map

**Q43. Yield-monitor data must be corrected for:**

(a) Flow (time) lag through the combine (b) Colour
(c) Wind chill (d) Fuel type

**Q63. Variable-rate seeding changes:**

(a) Only depth (b) Seed rate/spacing across the field
(c) Tyre pressure (d) PTO speed

**Answers**

Q41 — (a) Moisture, header width, speed and GPS position

Q42 — (b) Yield map

Q43 — (a) Flow (time) lag through the combine

Q63 — (b) Seed rate/spacing across the field

**Sources:** Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.